Data Structures (ETCCDS202)

Unit–2 Assignment: Linear Data Structures Toolkit (LDST)

Course Details

Name of the School: School of Engineering & Technology

Program: B.Tech CSE (AI ML)

Course Title: Data Structures

Course Code: ETCCDS202

Unit Title: Linear Data Structures

Student Name: Ankit Kumar Mishra

Roll No.: 2501730079

Section: A

Introduction

Data Structures is an important subject in computer science which helps us to store and organize data in an efficient way. In this unit, I studied about Linear Data Structures, where data elements are arranged in a sequence one after another.

These data structures are used in many real-life applications. For example, arrays are used to store data like marks, linked lists are used in playlists, stacks are used in undo operations, and queues are used in task scheduling.

In this assignment, I implemented different linear data structures using Python. I also learned how they work internally and how different operations are performed. This assignment helped me to improve my programming skills and understanding of data structures.

Dynamic Array

Description:

I implemented a dynamic array which increases its size automatically when it becomes full.

Operations:

- append(x) → insert element
- pop() → remove last element
- display → show elements

Amortized Time Complexity (Simple Explanation)

When we insert elements using append:

- Most of the time it takes O(1) time
- But when the array becomes full, resizing is required
- Resizing takes O(n) time because all elements are copied

But resizing does not happen every time.

So overall average time is O(1)
This is called Amortized Time Complexity

Linked List

(A) Singly Linked List

Operations performed:

- Insert at beginning
- Insert at end

- Delete by value
- Traverse (print elements)

Explanation:

In singly linked list, each node stores data and a pointer to the next node. It allows easy insertion and deletion without shifting elements.

(B) Doubly Linked List

Operations:

- Insert after a node
- Delete at a position

Explanation:

Each node contains:

- data
- next pointer
- previous pointer

This allows traversal in both directions.

Stack and Queue

(A) Stack (LIFO)

Stack follows Last In First Out principle.

Operations:

- Push
- Pop
- Peek

Time Complexity:

- Push → O(1)
- Pop → O(1)
- Peek → O(1)

(B) Queue (FIFO)

Queue follows First In First Out principle.

Operations:

- Enqueue
- Dequeue
- Front

Time Complexity:

- Enqueue → O(1)
- Dequeue → O(1)
- Front → O(1)

Balanced Parentheses Checker

Description:

I implemented a function to check whether a given expression has balanced parentheses or not using stack.

Logic:

- Push opening brackets into stack
- When closing bracket appears → check top
- If matched → pop
- Else → not balanced

Test Cases:

Input	Output
([])	True
([)]	False
(((	False
""	True

Learning Outcomes

From this assignment, I learned:

- Implementation of arrays and linked lists
- Difference between static and dynamic structures
- Working of stack and queue

- Time complexity in simple terms
- Solving real-world problems using stack

Conclusion

In this assignment, I implemented different linear data structures such as Dynamic Array, Singly Linked List, Doubly Linked List, Stack, and Queue. I also created an application of stack which is the Balanced Parentheses Checker.

Through this assignment, I understood how different data structures work and how they are used to solve problems. I learned that arrays are simple but have limitations, while linked lists are more flexible. I also understood the working of stack (LIFO) and queue (FIFO) clearly.

While working on this assignment, I faced some difficulty in understanding linked lists and pointers, but after practicing, I was able to implement them correctly. I also learned about time complexity and why some operations take more time than others.

Overall, this assignment improved my coding skills and helped me understand the basics of data structures in a practical way. It also increased my confidence in writing programs without using built-in functions.

This assignment was very helpful for building my fundamentals in Data Structures.